Shanice Reih H. Tanque

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Education

Mindanao State University - Iligan Institute of Technology — Iligan City, Lanao del Norte
Bachelor of Science in Computer Applications — 2023 - 2027
Major in Embedded Systems
GPA: 1.32

Mindanao State University at Naawan - Integrated Developmental School — Naawan, Misamis Oriental
Science, Technology, Engineering, and Mathematics — 2017 - 2023
With Honors

Experience

Lead Generation Assistant — April 2025 - December 2025
- Researched and organized prospective client leads for marketing and sales outreach
- Gathered and verified relevant company and contact information from online sources
- Maintained structured lead data to support outreach and follow-up activities

Research Extension Division, MSU at Naawan — Naawan, Misamis Oriental
Work Immersion Intern — January 2023 - April 2023
- Assisted with water-quality and abnormality testing for samples from Panguil Bay
- Supported species identification and counting through microscope-based analysis
- Assisted research personnel with laboratory procedures and research-related data gathering

Projects

Development of YOLOv8n Aphid Damage Detection and Offline Monitoring for Bell Pepper (Capsicum annuum) on Raspberry Pi 5 (In Progress)
- Developing an offline Edge-IoT system using YOLOv8n on Raspberry Pi 5 to detect visible aphid-induced damage in bell pepper leaves under field conditions
- Designing localized image-based detection and dashboard monitoring with planned evaluation of detection performance, inference speed, power consumption, offline operation, and usability
- **Research Page:**Project Research Page

RT-Spark Personal Music Player
- Developed and physically verified a real-time embedded music player on the STM32F407ZG RT-Spark board using FreeRTOS, STM32Cube, PlatformIO, ES8388 audio, I2S3, and DMA
- Designed a three-task architecture for eight-song selection, playback control, ADC potentiometer volume, ST7789 LCD/RGB status output, and mutex-protected resources; validated with 14 host test suites and nine build environments
- **Link:**GitHub

FreeRTOS Multisensor Room Monitoring System
- Developed an STM32F103C8T6 room-monitoring system with five FreeRTOS tasks using queues, task notifications, and a recursive mutex for sensing, display, input, alarms, and motion handling
- Integrated DHT22, LDR, PIR, SSD1306 OLED, rotary encoder, and PWM buzzer peripherals; verified functionality through 13 unit tests and 10 Wokwi functional tests
- **Links:**GitHub | Hackster

Industrial Drying Oven PLC-HMI Control System
- Designed and simulated a thermal-control system in CODESYS using IEC 61131-3 Structured Text and a five-state finite-state machine for Idle, Heating, Drying, Cooling, and Alarm/Shutdown operation
- Implemented HMI setpoints and status displays, automatic door interlocks, a 180°C warning, 200°C emergency shutdown, 25°C safe-open threshold, and hysteresis-based heater control; verified behavior through simulation
- **Link:**Project Site

VERTANOM - Smart Vertical Farming System
- Co-developed a Qt/C++ smart vertical farming dashboard for soil moisture, pH, temperature, and humidity monitoring using Qt SQL and SQLite
- Implemented sensor-data visualization and latest-value/status displays; project recognized as Champion at the Sustainable Innovation Pitch 2024
- **Link:**GitHub

Technical Skills

Programming Languages: C, C++, Structured Text (IEC 61131-3), SQL
Embedded Systems & IoT: STM32, FreeRTOS, RT-Thread, Raspberry Pi 5, Arduino, ESP32, Sensor Integration, Edge IoT
Industrial Automation: CODESYS, PLC Programming, HMI Development, Finite State Machines, Safety Interlocks
Frameworks & Tools: STM32Cube HAL, Qt, LVGL, PlatformIO, Visual Studio Code, Arduino IDE, Wokwi, Git, GitHub
Hardware & Interfaces: I2S, DMA, ADC, PWM, I2C, OLED/LCD, ES8388 Audio Codec, Sensors and Actuators
Databases & Testing: SQLite, MySQL, Unity Test Framework, Cppcheck, Host and Functional Testing

Leadership & Activities

College of Computer Studies - Student Council, MSU-IIT — Iligan City, Lanao del Norte
Undersecretary, Ways and Means Committee — August 2025 - May 2026

- Supported council initiatives involving resource planning, coordination, and organizational activities
- Contributed to student council projects and activities throughout the academic year
- Received multiple Special Awards of Excellence recognizing leadership, governance, collaboration, impact, and stewardship

College of Computer Studies - Student Council, MSU-IIT — Iligan City, Lanao del Norte
Co-Chairperson, Culture and Arts Committee — August 2024 - May 2025

- Co-led Culture and Arts Committee initiatives and supported the planning and implementation of college events
- Coordinated with fellow council members in organizing activities for the College of Computer Studies
- Received a Certificate of Recognition for commendable service as Culture and Arts Committee Co-Chairperson

myComApps Technology Symposium & Exhibit - RT-Thread-Based Technology — Iligan City, Lanao del Norte
Overall Committee Coordinator & Attendance Committee Member — November 2025

- Coordinated committee leads and supported overall event operations across the symposium and exhibit
- Handled attendance-related responsibilities while also participating as a project exhibitor
- Developed and presented BrainBoard, an RT-Thread and LVGL-based interactive embedded GUI project

my.ComApps Technology Development Program — Iligan City, Lanao del Norte
Head, Attendance and Certificate Committee — December 2024

- Led the Attendance and Certificate Committee for the technology development program
- Coordinated participant attendance records and certificate preparation and distribution
- Worked with other committees to support smooth program operations

Achievements

Sustainable Innovation Pitch 2024 - Champion — Iligan City, Lanao del Norte
MSU-IIT Sustainable Development Day — September 2024

- Won the Sustainable Innovation Pitch during the 2024 Sustainable Development Day Celebration for VERTANOM
- Recognized for developing an innovative smart vertical farming solution

TechShowcase 2025 - IoT System for Aphids Damage Detection in Bell Pepper Farms — Iligan City, Lanao del Norte
MSU-IIT — December 2025

- Best in Pitching - Champion
- Best in Innovative System - 1st Runner-Up
- Presented an IoT-based system for agricultural pest-damage detection during TechShowcase 2025

Certificate of Commendation - CCC102 Computer Programming 2 — Iligan City, Lanao del Norte
College of Computer Studies, MSU-IIT — March 2025

- Recognized for creative technological works, inspirational presentations, and contributions that supported students preparing for final project milestones
- Commended for dedication to education, mentorship, and fostering student growth and excellence

Certifications & Training

Startup Basics 101 - iDEYA:CIT, MSU-IIT — September 2026

- Completed a one-day learning session on startups, innovation, and entrepreneurship

AI Career Readiness Training - ASEAN Ahead, ASEAN Foundation — August 2026

- Completed the AI Career Readiness Training through ASEAN Ahead, organized by the ASEAN Foundation